

## EXAMINATION 2 | CHEM 2313 | FALL 2019

Write your final answers in the boxes provided below. Additional paper has been provided at the end of the exam for scratch work along with supplemental tables. Only pencil, paper, and a model kit are permitted. Write all answers legibly and draw all structures clearly with sensible bond angles.

Name

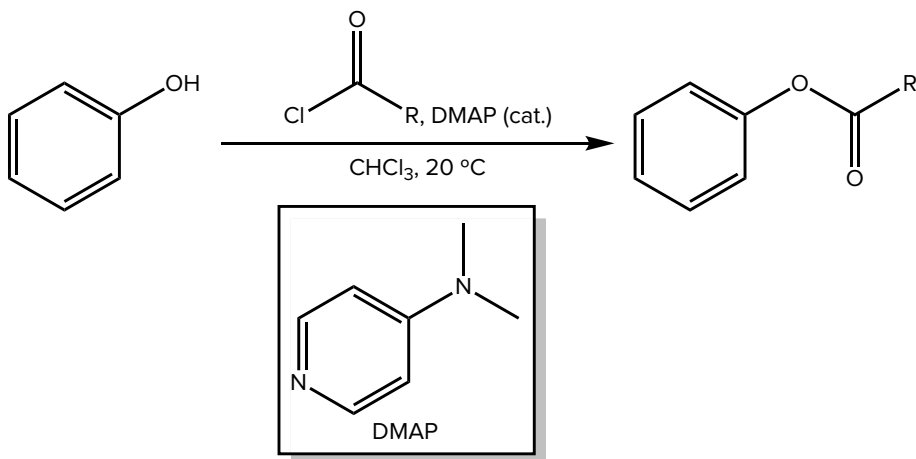
GT ID Number

| <b>Section</b>               | <b>Total Points</b> |
|------------------------------|---------------------|
| I. Multiple-choice Questions | 20                  |
| II. Predict the Products     | 20                  |
| III. Short-answer Questions  | 30                  |
| IV. Synthetic Planning       | 10                  |
| V. Reaction Mechanism        | 20                  |

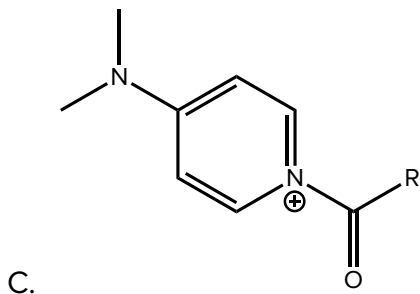
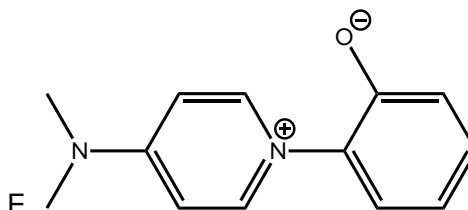
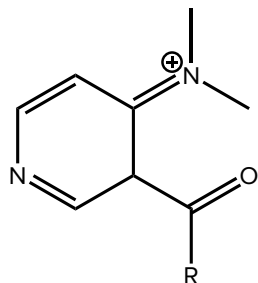
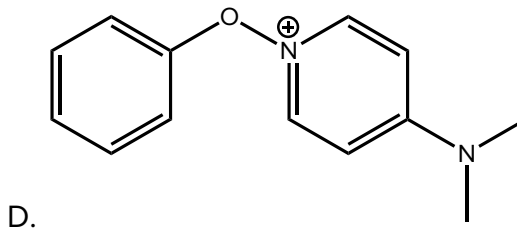
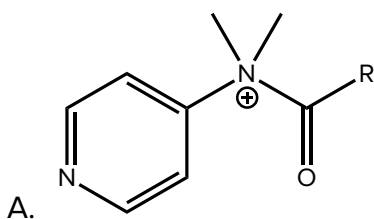
### I. Multiple-Choice Questions

Choose the best answer among the choices provided. Circle your choice *and* write its letter in the box next to each question.

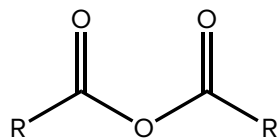
1. *N,N*-Dimethylaminopyridine (DMAP) catalyzes acylation reactions of various nucleophiles. It acts by activating the acylating agent. An example is shown below.



DMAP is a Lewis basic (nucleophilic) catalyst. Which of the following intermediates is most likely formed during the mechanism of this reaction?



2. Which of the following nucleophiles will **NOT** react favorably with an acid anhydride in a nucleophilic acyl substitution reaction?

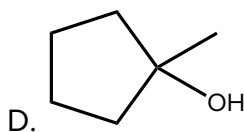
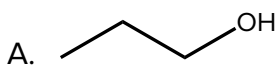
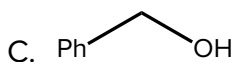
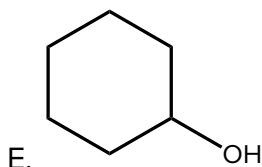
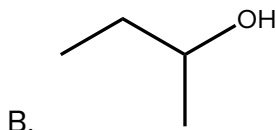


acid anhydride

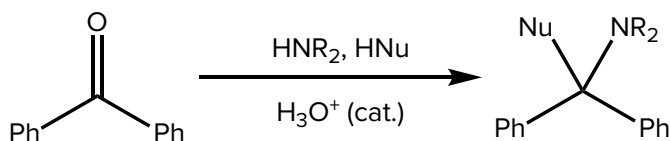
☐

- A. Sulfonate ( $\text{RO}_2\text{SO}^-$ )
- B. Hydroxide ( $\text{HO}^-$ )
- C. Thiohydroxide ( $\text{HS}^-$ )
- D. Amine ( $\text{HNR}_2$ )
- E. Amide anion ( $\text{R}_2\text{N}^-$ )

3. Which of the following alcohols does not react in the presence of aqueous chromic acid ( $\text{H}_2\text{CrO}_4$ ,  $\text{H}_3\text{O}^+$ )?

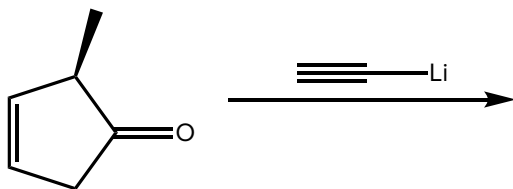

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4. Treatment of benzophenone with a secondary amine results in products with the general structure shown below. What kind of intermediate is most likely involved in the mechanism of this reaction?


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- A. Enol
- B. Enolate
- C. Enamine
- D. Imine
- E. Iminium ion

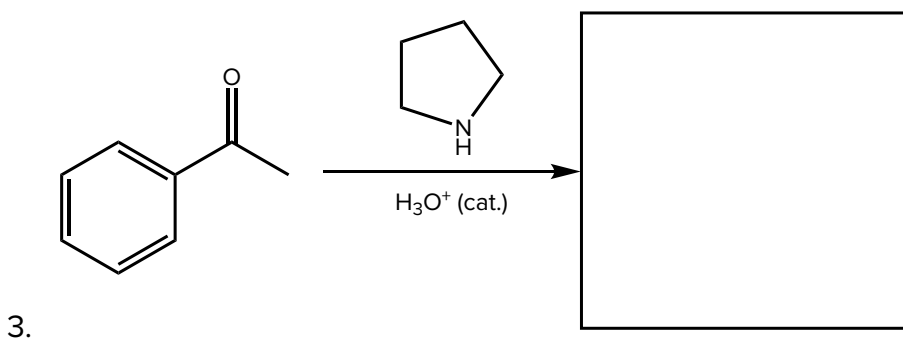
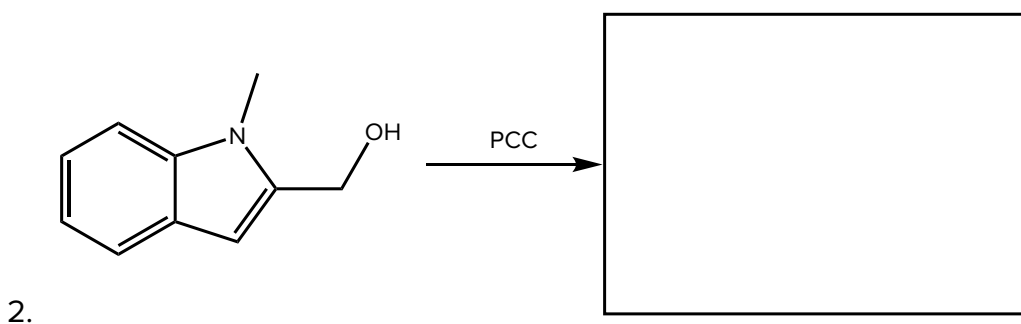
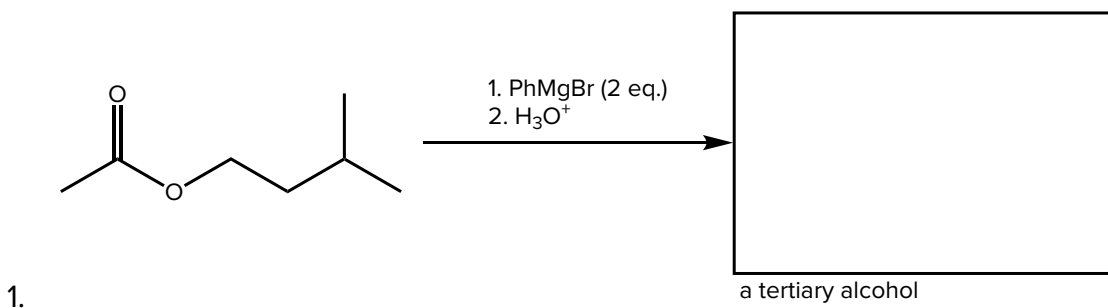
5. When ethynyllithium ( $\text{HCCLi}$ ) reacts with the ketone shown below, two products are possible. What is the isomeric relationship between the products?


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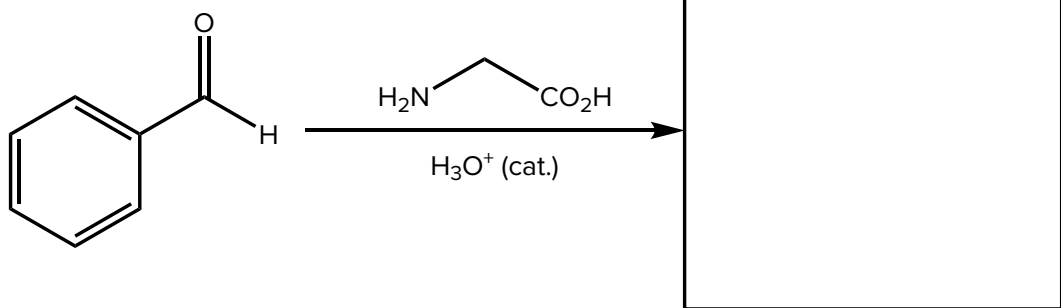
- A. Homomers (i.e., the “two products” are in fact identical)  
 B. Enantiomers  
 C. Diastereomers  
 D. Constitutional isomers  
 E. Polymers

## II. Predict the Products

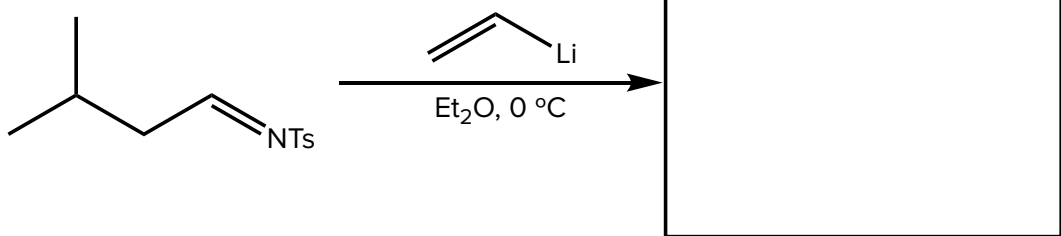
Provide the major neutral organic product for each reaction or sequence of reactions below after appropriate workup. Clearly indicate the configurations of all stereocenters (except for those with random configurations; stereochemistry can be omitted at these).



4.



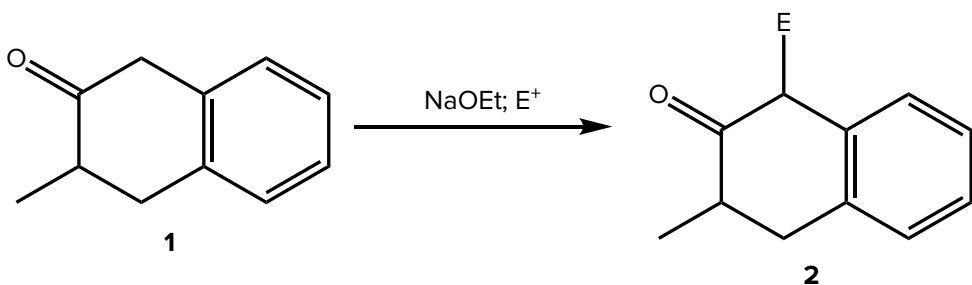
5.



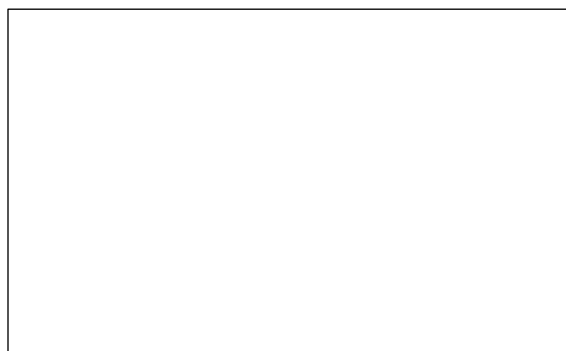
### III. Short-answer Questions

For each question below, provide the requested structure, mechanism, or explanation. Ensure that your complete response fits within the box provided.

- Reaction of ketone **1** with sodium ethoxide followed by treatment with an electrophile  $\text{E}^+$  led *exclusively* to product **2**.

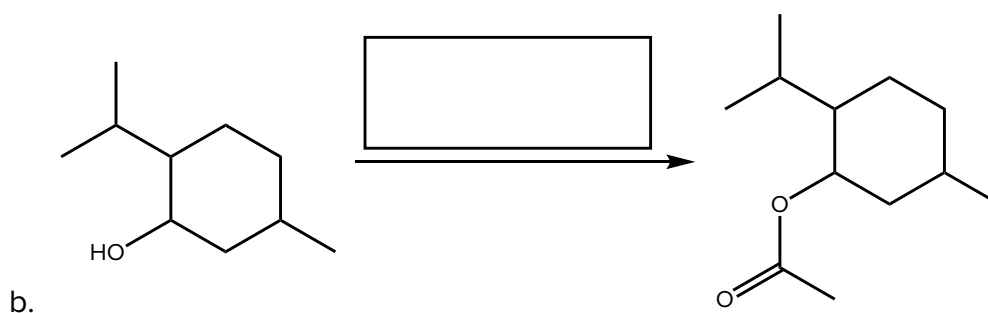
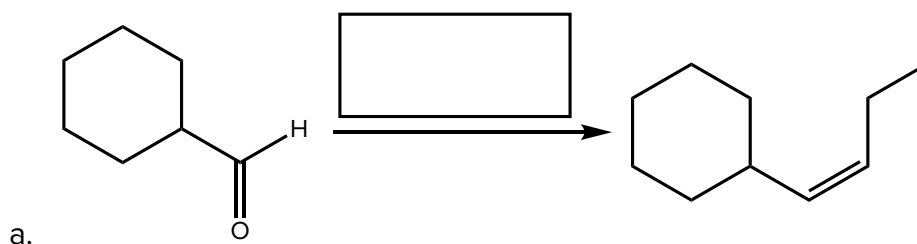


- Draw the structure of another product that could form under these conditions. We will call this product **3**.



- b. Explain why **2** forms in preference to **3**. Refer to the key intermediate of these reactions and one of the five stability factors in your response.

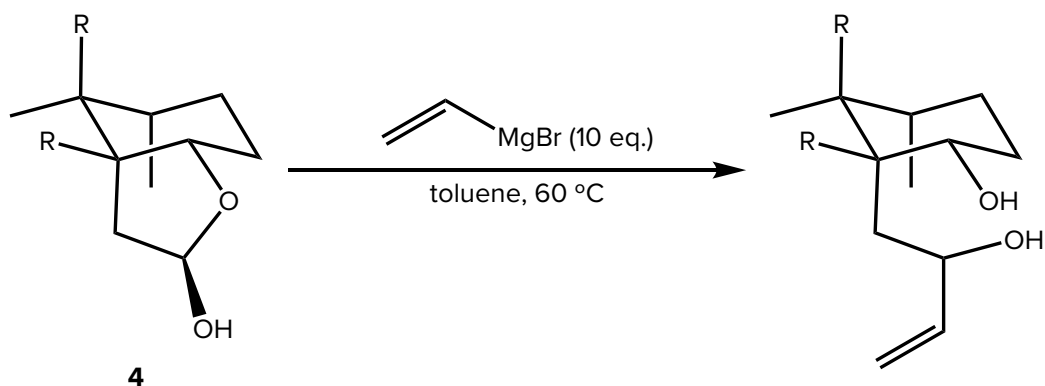
2. In the box above each arrow in the reaction schemes below, provide a set of reagents that will accomplish the transformation.



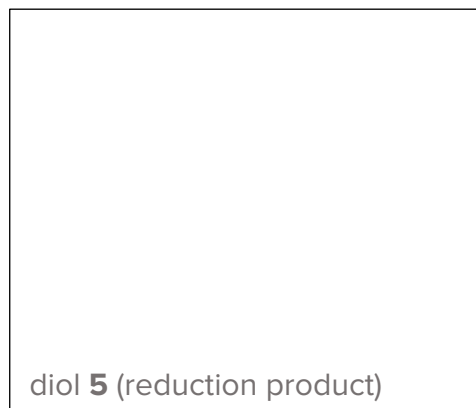
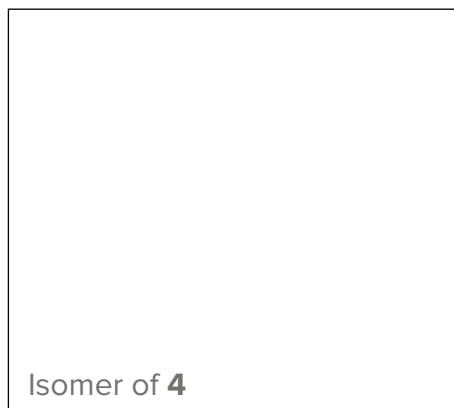
3. In the box below, draw the structure of a ketone or aldehyde whose enol form is comparable in stability to ( $K \approx 1$ ) or greater in stability than ( $K > 1$ ) its keto form.

4. In the box at right, draw the structure of a ketone or aldehyde that will be hydrated to a negligible degree ( $K < 10^{-5}$ ) when it is dissolved in water.

5. The reaction below occurs in the midst of a synthesis of a complex natural product. This reaction is remarkable because a hemiacetal, ordinarily resistant to reaction with nucleophiles, reacts with vinyl magnesium bromide.



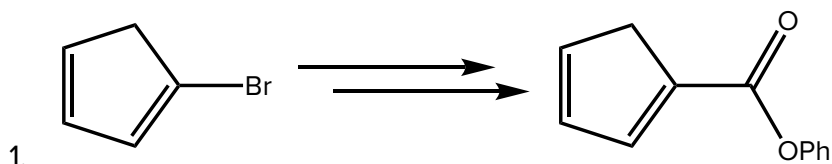
- a. Starting material **4** is in equilibrium with an isomer containing a carbonyl group. Draw the structure of the isomer in the box below.



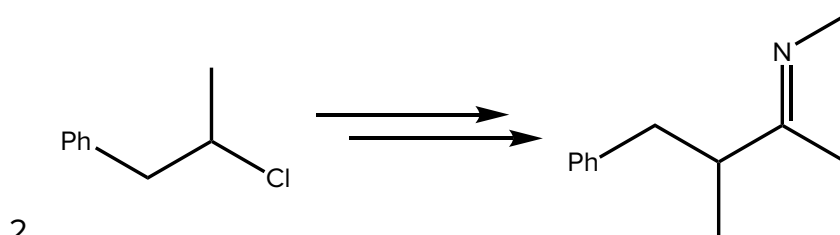
- b. The authors of this synthesis treated **4** with lithium aluminum hydride and observed slow reduction to a diol product **5**. Draw the structure of this product in the box above.

#### IV. Synthetic Planning

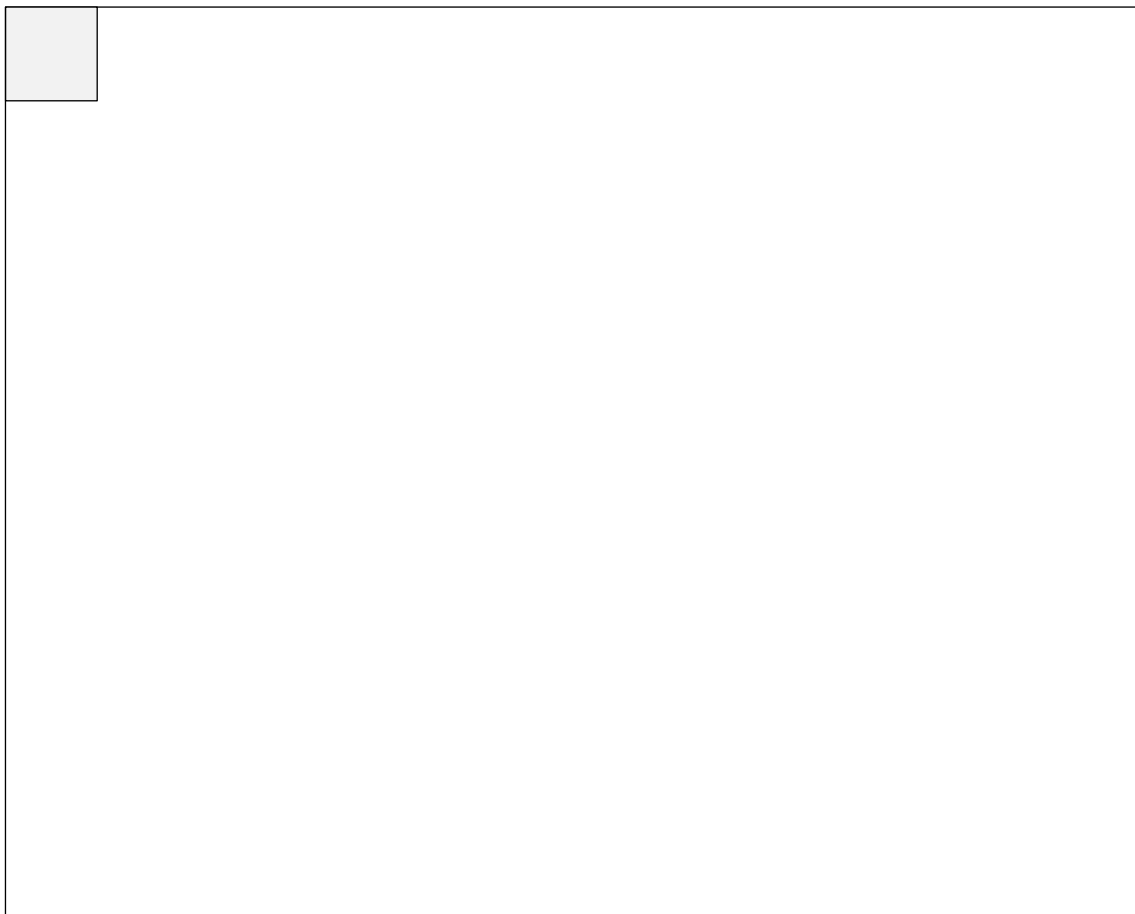
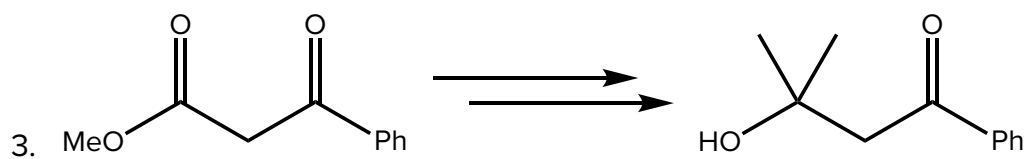
Of the three problems below, **choose two** and provide a sequence of reaction conditions that will accomplish the desired transformation. Include all isolated compounds, but do not draw curved arrows or any reactive intermediates. **Place an X in the shaded box next to the two problems you would like graded.**



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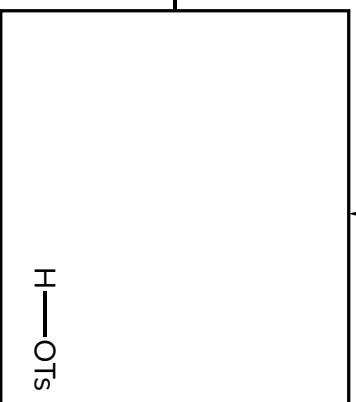
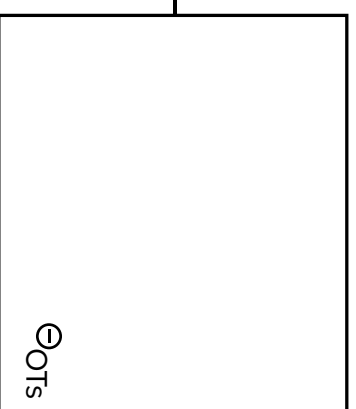
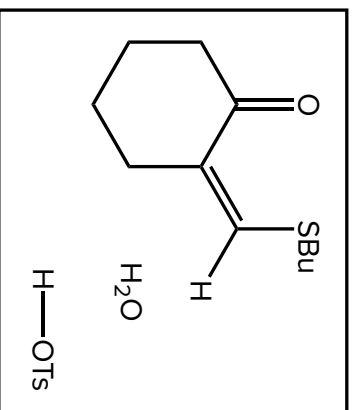
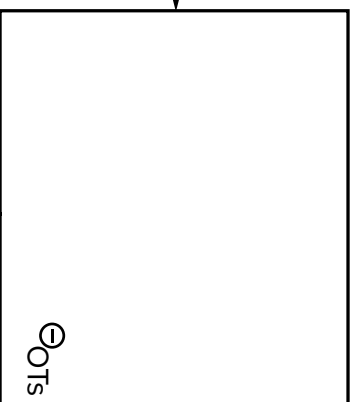
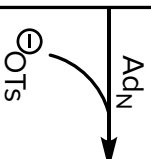
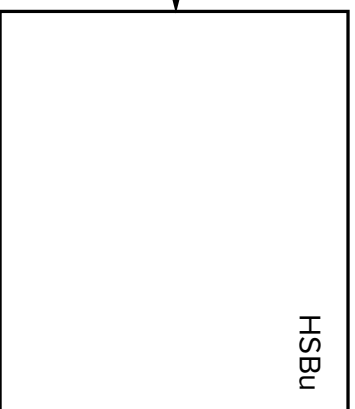
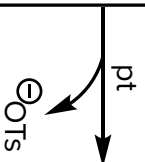
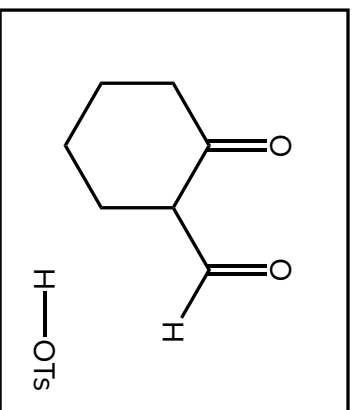
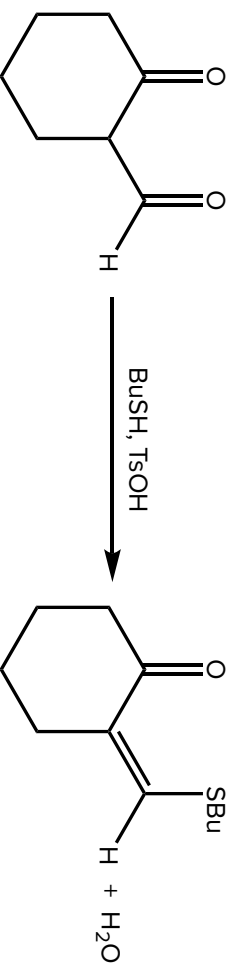


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### V. Reaction Mechanism

To complete the reaction mechanism below, add curved arrows to the reactants of each elementary step and draw the products in the boxes provided. Each box should include a set of reactants and curved arrows except for the last, which will only include products. Note the step labels provided.



# pKa Chart

1

conjugate acid

conjugate base

sulfuric acid



-10

strongest acids

hydroiodic acid



-9

weakest bases

hydrobromic acid



-8

protonated ketone



-7.3

hydrochloric acid



-7

protonated ether



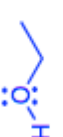
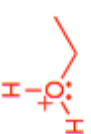
-3.6

carbocations



-3

protonated alcohol



-2.4

hydronium ion



-1.7

2

conjugate acid

conjugate base

nitric acid



-1.3

tosic acid



-0.6

hydrofluoric acid



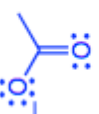
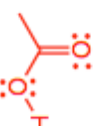
3.2

hydrogen nitride



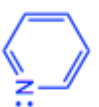
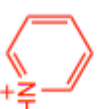
4.7

carboxylic acids



4.8

protonated pyridine



5.2

carbonic acid



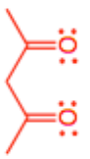
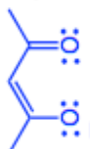
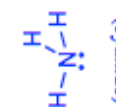
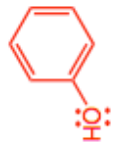
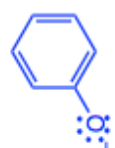
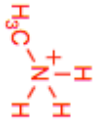
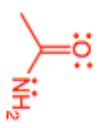
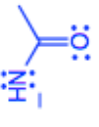
6.37

hydrogen sulfide

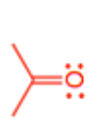
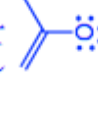
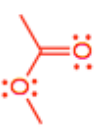
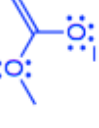
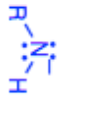
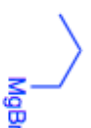


7

3

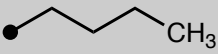
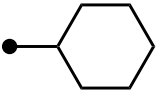
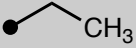
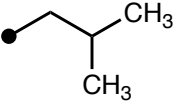
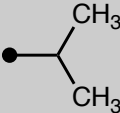
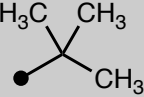
| conjugate acid                                                                                         | conjugate base                                                                                        |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1,3-diketones<br>   | <br>9              |
| hydrogen cyanide<br>$\text{H}-\text{C}\equiv\text{N}:$                                                 | $:\text{C}\equiv\text{N}^-:$<br>(cyanide)<br>9.1                                                      |
| ammonium ions<br>    | <br>9.4             |
| phenols<br>           | <br>10               |
| bicarbonate<br>$\text{HCO}_3^-$                                                                        | $\text{CO}_3^{2-}$<br>10.25                                                                           |
| thiols<br>            | <br>10.5             |
| protonated amines<br> | <br>10.6             |
| cyclopentadiene<br>   | <br>(aromatic)<br>15 |
| amides<br>            | <br>15               |

4

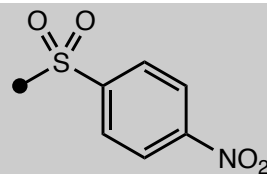
| conjugate acid                                                                                            | conjugate base                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| water<br>            | $:\ddot{\text{O}}-\text{H}^-$<br>(hydroxide)<br>15.7                                                                              |
| primary alcohols<br> | <br>(alkoxides)<br>16                        |
| tertiary alcohols<br> | <br>(tert-butoxide)<br>18                     |
| ketones<br>            | <br>(enolates)<br>18-21                        |
| esters<br>             | <br>(ester enolates)<br>25                     |
| alkynes<br>$\text{C}\equiv\text{C}-\text{H}$                                                              | $\text{C}\equiv\text{C}:^-$<br>(acetylide anions)<br>26                                                                           |
| hydrogen<br>$\text{H}-\text{H}$<br>one way only                                                           | $:\text{H}^-$ (hydride)<br>35                                                                                                     |
| ammonia/amines<br>     | <br>(amide bases)<br>36                        |
| alkanes<br>            | <br>~60<br>(Grignards/ organolithium reagents) |
| weakest acids                                                                                             | strongest bases                                                                                                                   |

## Abbreviations in Organic Chemical Structures

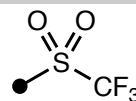
### Alkyl Groups

| Abbreviation | Structure                                                                         |
|--------------|-----------------------------------------------------------------------------------|
| Bu           |  |
| Cy           |  |
| Et           |  |
| i-Bu         |  |
| i-Pr         |  |
| Me           |  |
| t-Bu         |  |

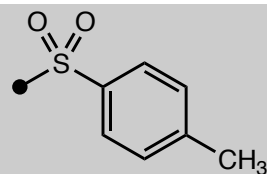
Ns



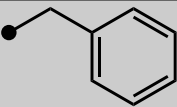
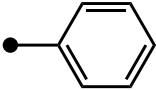
Tf



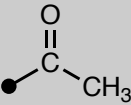
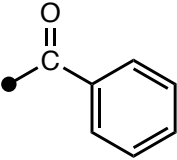
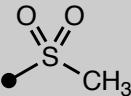
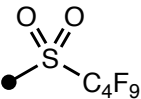
Ts



### Arenes

| Abbreviation | Structure                                                                           |
|--------------|-------------------------------------------------------------------------------------|
| Bn           |  |
| Ph           |  |

### Heteroatom-containing Groups

| Abbreviation | Structure                                                                           |
|--------------|-------------------------------------------------------------------------------------|
| Ac           |  |
| Bz           |  |
| Ms           |  |
| Nf           |  |

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